

Alex-Ciprian Cîmpan

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Education

Technical University of Cluj-Napoca – BS in Computer Engineering

Expected Jul 2027

Relevant Coursework – Operating Systems, Computer Architecture, Data Structures, Fundamental Algorithms, Image Processing.

Skills

Programming: C/C++, Java, Python, SQL, Assembly

Tools: NumPy, pandas, OpenCV, Git

Soft Skills: teamwork, adaptability, attention to detail, problem-solving

Projects

Nutrition Label Analyzer

Python, EasyOCR, Streamlit

github.com/alexcipri21/nutrition-label-analyzer

- Developed a multi-agent pipeline application designed to scan and interpret macronutrients from product labels.
- Integrated a local EasyOCR visual engine for robust text extraction from image sources.
- Orchestrated an asynchronous dual-agent analysis system utilizing Qwen2.5-7B via Hugging Face API.

Real-Time Parking Spot Detection System

Python, PyTorch, OpenCV, Deep Learning

github.com/alexcipri21/parking-spot-detector

- Built a real-time parking monitoring system integrating classic image processing and Deep Learning.
- Solved camera shake issues by implementing ECC stabilization to keep parking spot boundaries perfectly aligned.
- Optimized computational efficiency by skipping redundant checks on occupied spots, updating empty spots more frequently.

Collaborative Bug Tracking Platform

React, Spring Boot, Cypress, GitHub Actions

github.com/marcuandrei1/bug-tracker

- **Team Project:** Co-developed a full-stack bug tracking system, focusing on testing, CI/CD, and user moderation.
- Implemented Unit and Integration tests in Java, alongside end-to-end (E2E) testing using Cypress.
- Automated testing pipelines using GitHub Actions for continuous integration (CI/CD).
- Developed a user-banning module with automated notifications via Email (Spring Mail) and WhatsApp Web API.

RideSharing Platform

React, Node.js, Google Maps API

- Designed system architecture for a ride-sharing application with focus on backend flow and trip management.
- Built React-based frontend and integrated Google Maps API for location and routing features.
- Explored REST API design and frontend-backend communication patterns.

Performance Tester

Java, Swing, Multithreading

github.com/alexcipri21/performance-tester

- Developed a benchmarking tool for CPU, memory, and disk performance measurement.
- Implemented multithreaded stress tests and JSON-based result storage.